ACE Smart-Grid" — Decoupled Automation System

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UNIT – II
MINI PROJECT

```
//Mini-Project Name: "ACE Smart-Grid" — Decoupled Automation System " -
// Register smart devices
// Turn devices ON/OFF
// Display device status
// Monitor power consumption
// Generate total energy usage report

package miniprojects;

import java.util.ArrayList;
import java.util.Scanner;

interface SmartDevice {
    void turnOn();
    void turnOff();
    void displayStatus();
    int getPowerUsage();
}

class SmartLight implements SmartDevice {

    private int id;
    private boolean status;

    SmartLight(int id) {
        this.id = id;
        status = false;
    }
```

```
    public void turnOn() {
        status = true;
    }

    public void turnOff() {
        status = false;
    }

    public void displayStatus() {
        System.out.println("Light ID : " + id +
                " | Status : " +
                (status ? "ON" : "OFF"));
    }

    public int getPowerUsage() {
        return status ? 20 : 0;
    }
}

class SmartFan implements SmartDevice {

    private int id;
    private boolean status;

    SmartFan(int id) {
        this.id = id;
        status = false;
    }

    public void turnOn() {
```

```
        status = true;
    }

    public void turnOff() {
        status = false;
    }

    public void displayStatus() {
        System.out.println("Fan ID : " + id +
                " | Status : " +
                (status ? "ON" : "OFF"));
    }

    public int getPowerUsage() {
        return status ? 75 : 0;
    }
}

public class SmartGridAutomation {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        ArrayList<SmartDevice> devices = new ArrayList<>();

        devices.add(new SmartLight(101));
        devices.add(new SmartFan(201));

        int choice;
```

```
    do {

        System.out.println("\n===== SMART GRID SYSTEM
=====");
        System.out.println("1. Turn ON All Devices");
        System.out.println("2. Turn OFF All Devices");
        System.out.println("3. View Device Status");
        System.out.println("4. Total Power Usage");
        System.out.println("5. Exit");

        System.out.print("Enter Choice : ");
        choice = sc.nextInt();

        switch (choice) {

            case 1:

                for (SmartDevice d : devices)
                    d.turnOn();

                System.out.println("All Devices Turned ON");
                break;

            case 2:

                for (SmartDevice d : devices)
                    d.turnOff();

                System.out.println("All Devices Turned OFF");
                break;
```

```
            case 3:
                for (SmartDevice d : devices)
                    d.displayStatus();
                break;

            case 4:
                int total = 0;
                for (SmartDevice d : devices)
                    total += d.getPowerUsage();
                System.out.println("Total Power Usage : "
                        + total + " Watts");
                break;

            case 5:
                System.out.println("Thank You");
                break;
            default:
                System.out.println("Invalid Choice");
            }

        } while (choice != 5);

        sc.close();
    }
}
```

OUTPUT:-

===== SMART GRID SYSTEM =====
1. Turn ON All Devices
2. Turn OFF All Devices
3. View Device Status
4. Total Power Usage
5. Exit

Enter Choice : 1
All Devices Turned ON

===== SMART GRID SYSTEM =====
1. Turn ON All Devices
2. Turn OFF All Devices
3. View Device Status
4. Total Power Usage
5. Exit

Enter Choice : 3
Light ID : 101 | Status : ON
Fan ID : 201 | Status : ON

===== SMART GRID SYSTEM =====
1. Turn ON All Devices
2. Turn OFF All Devices
3. View Device Status
4. Total Power Usage
5. Exit

Enter Choice : 4
Total Power Usage : 95 Watts

===== SMART GRID SYSTEM =====
1. Turn ON All Devices
2. Turn OFF All Devices
3. View Device Status
4. Total Power Usage
5. Exit
Enter Choice : 2
All Devices Turned OFF

===== SMART GRID SYSTEM =====
1. Turn ON All Devices
2. Turn OFF All Devices
3. View Device Status
4. Total Power Usage
5. Exit
Enter Choice : 3
Light ID : 101 | Status : OFF
Fan ID : 201 | Status : OFF

===== SMART GRID SYSTEM =====
1. Turn ON All Devices
2. Turn OFF All Devices
3. View Device Status
4. Total Power Usage
5. Exit
Enter Choice :